Morocco NDVI Analysis

using Power BI and Google Earth Engine

Vegetation Analysis Report — 2010 to 2024

Satellite data: MODIS / VIIRS Climate data: CHIRPS, ERA5, Copernicus S5P

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I. Introduction

Vegetation dynamics represent an essential indicator for understanding the ecological state of a territory and its evolution in response to climatic factors. In Morocco, a country characterised by high climatic variability and growing pressure on natural resources, analysing vegetation dynamics is particularly strategic. The Normalized Difference Vegetation Index (NDVI) is today one of the most reliable indices for measuring the density, vigour, and health of vegetation at large scales.

This report analyses the evolution of NDVI in Morocco between **2010 and 2024**, drawing on satellite data from VIIRS and MODIS sensors, combined with climatic factors such as precipitation, luminosity, temperature, and CO_2 emissions. The objective is to identify the spatio-temporal trends of vegetation and determine which factors most significantly influence NDVI dynamics.

The study is based on a two-pronged approach:

1. A **descriptive exploration** of NDVI evolution by year, season, and region;
2. A **correlational analysis** assessing the impact of climatic parameters on vegetation cover.

II. Data Extraction (Google Earth Engine)

1. Data Sources

Data is extracted from Google Earth Engine (GEE) using Python. The following image collections are used:

Variable	GEE Collection	Aggregation
NDVI	`MODIS/061/MOD13A2`	Monthly mean
Precipitation	`UCSB-CHG/CHIRPS/DAILY`	Monthly sum (mm)
Temperature	`ECMWF/ERA5_LAND/MONTHLY`	Monthly mean (°C)
CO_2	`COPERNICUS/S5P/OFFL/L3_CO`	Monthly mean
Luminosity	`NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG`	Monthly mean

Administrative boundaries are taken from `FAO/GAUL_SIMPLIFIED_500m/2015/level1`. Values are extracted at **1 km** resolution and reduced per region using `ee.Reducer.mean()`.

2. Extraction Code

```python
import ee
import geemap.core as geemap

ee.Authenticate()
ee.Initialize(project="shining-glyph-466811-u6")
```

Listing 1: GEE initialisation

```python
COUNTRY    = 'Morocco'
START_YEAR = 2010
END_YEAR   = 2024
SCALE_M    = 1000   # ~1 km

adm1 = (ee.FeatureCollection('FAO/GAUL_SIMPLIFIED_500m/2015/level1')
          .filter(ee.Filter.eq('ADM0_NAME', COUNTRY))
          .select(['ADM0_NAME', 'ADM1_NAME']))

ic_ndvi   = ee.ImageCollection('MODIS/061/MOD13A2').select('NDVI')
ic_chirps = ee.ImageCollection('UCSB-CHG/CHIRPS/DAILY').select('precipitation')
ic_era5m  = ee.ImageCollection('ECMWF/ERA5_LAND/MONTHLY').select('temperature_2m')
```

```
ic_co     = (ee.ImageCollection('COPERNICUS/S5P/OFFL/L3_CO')
               .select('CO_column_number_density'))
ic_viirs  = ee.ImageCollection('NOAA/VIIRS/DNB/MONTHLY_V1/VCMCFG')
   .select('avg_rad')
```

Listing 2: Country, region, and dataset definitions

```
years  = ee.List.sequence(START_YEAR, END_YEAR)
months = ee.List.sequence(1, 12)

def masked_band(name):
    return ee.Image.constant(0).updateMask(ee.Image.constant(0)).
       rename(name)

def monthly_panel(y, m):
    y = ee.Number(y); m = ee.Number(m)
    start = ee.Date.fromYMD(y, m, 1)
    end   = start.advance(1, 'month')

    def safe_mean(col, name):
        return ee.Image(ee.Algorithms.If(
            col.filterDate(start, end).size().gt(0),
            col.filterDate(start, end).mean().rename(name),
            masked_band(name)))

    def safe_sum(col, name):
        return ee.Image(ee.Algorithms.If(
            col.filterDate(start, end).size().gt(0),
            col.filterDate(start, end).sum().rename(name),
            masked_band(name)))

    ndvi   = safe_mean(ic_ndvi,   'ndvi').multiply(0.0001)
    precip = safe_sum (ic_chirps, 'precip_mm')
    t2m    = safe_mean(ic_era5m,  't2m_c').subtract(273.15)
    co     = safe_mean(ic_co,     'co_col_molm2')
    viirs  = safe_mean(ic_viirs,  'viirs_rad')

    cube = ee.Image.cat([ndvi, precip, t2m, co, viirs])

    stats = (cube.reduceRegions(
                 collection=adm1,
                 reducer=ee.Reducer.mean(),
                 scale=SCALE_M,
                 tileScale=4)
             .map(lambda f: f.set({'year': y, 'month': m})))
    return stats
```

Listing 3: Monthly aggregation per region

```
ym_list = years.map(
    lambda y: months.map(
        lambda m: ee.Dictionary({'y': y, 'm': m}))).flatten()

all_stats = ee.FeatureCollection(
    ym_list.map(lambda d: monthly_panel(
        ee.Dictionary(d).get('y'),
        ee.Dictionary(d).get('m')))).flatten()

```

```
task = ee.batch.Export.table.toDrive(
    collection     = all_stats,
    description    = f"{COUNTRY}_NDVI_Rain_Temp_2010_2024",
    folder         = 'EarthEngineExports',
    fileNamePrefix = f"{COUNTRY}_NDVI_Rain_Temp_2010_2024",
    fileFormat     = 'CSV',
    selectors      = ['ADM0_NAME', 'ADM1_NAME', 'year', 'month',
                      'ndvi', 'precip_mm', 't2m_c',
                      'co_col_molm2', 'viirs_rad']
)
task.start()
```

Listing 4: CSV export to Google Drive

3. Dataset Overview

	A	B	C	D	E	F	G	H	I
1	ADM0_NAME	ADM1_NAME	year	month	ndvi	precip_mm	t2m_c	co_col_molm2	viirs_rad
2	Morocco	Chaouia - Ouardigha	2010	1	0.3875716128	112.3166017	11.0339174		
3	Morocco	Doukkala - Abda	2010	1	0.4095572862	66.68444833	12.80074361		
4	Morocco	Fès - Boulemane	2010	1	0.2426815966	100.2618795	7.025555927		
5	Morocco	Gharb - Chrarda - Bé	2010	1	0.4787396975	186.1467274	12.29668471		
6	Morocco	Grand Casablanca	2010	1	0.4469561425	125.0082508	13.06011696		
7	Morocco	Guelmim - Es-Semar	2010	1	0.1058451635	22.85802175	14.5531988		
8	Morocco	Laâyoune - Boujdou	2010	1	0.09735114819	40.76581981	16.72565102		
9	Morocco	Marrakech - Tensift -	2010	1	0.2478896721	51.89105281	10.15429029		
10	Morocco	Rabat - Salé - Zemm	2010	1	0.4816533698	135.1040879	10.84670993		
11	Morocco	Souss - Massa - Draâ	2010	1	0.163402689	51.13375982	9.941944554		
12	Morocco	Tadla - Azilal	2010	1	0.3376850135	124.5512982	5.618135784		
13	Morocco	Tanger - Tétouan	2010	1	0.5396675766	263.4220584	10.46116034		
14	Morocco	Taza - Al Hoceima -	2010	1	0.3125800603	160.1405722	8.984774583		
15	Morocco	Meknès - Tafilalet	2010	1	0.2056487385	67.14478516	8.532564592		
16	Morocco	Oriental	2010	1	0.1908965374	59.86949179	8.598850306		
17	Morocco	Chaouia - Ouardigha	2010	2	0.545024687	122.5386087	13.2213228		
18	Morocco	Doukkala - Abda	2010	2	0.5650209571	109.225271	14.75610573		
19	Morocco	Fès - Boulemane	2010	2	0.2661351339	92.85074821	9.577821842		
20	Morocco	Gharb - Chrarda - Bé	2010	2	0.5249219134	200.1983049	14.19830213		
21	Morocco	Grand Casablanca	2010	2	0.5660957566	104.4146941	14.9234962		
22	Morocco	Guelmim - Es-Semar	2010	2	0.1132850089	40.80360109	16.46672685		
23	Morocco	Laâyoune - Boujdou	2010	2	0.0976486144	46.37479212	18.35474606		
24	Morocco	Marrakech - Tensift -	2010	2	0.3462290844	120.6940948	12.4976284		
25	Morocco	Rabat - Salé - Zemm	2010	2	0.5761128026	143.0920218	13.10681884		
26	Morocco	Souss - Massa - Draâ	2010	2	0.1960245031	100.055441	11.72245492		
27	Morocco	Tadla - Azilal	2010	2	0.3900502217	166.7667785	8.453375379		

Figure 1: Preview of `Morocco_NDVI_Rain_Temp_2010_2024.csv`.

Notes on data availability:

- **Luminosity** (VIIRS): available from **July 2011** onwards only.
- **CO_2** (Copernicus S5P): available from **2018** onwards only.

III. Data Cleaning

After loading the CSV into **Power BI**, the *Transform Data* function is used to set column types and create the following computed variables:

- **Date:** combines month and year into a single date field.
- **Season:** classifies each record into Winter, Spring, Summer, or Autumn.
- **Variation:** $\Delta\mathrm{NDVI}_K = \overline{\mathrm{NDVI}}_K - \overline{\mathrm{NDVI}}_{K-1}$
- **Variation Category:** *Increase* if $\Delta > 0$, *Decrease* if $\Delta < 0$, *None* if $\Delta = 0$.

Region	Date	ndvi	Precipitation (mm)	Temperature (°C)	year	month	Saison	Emmision CO2	luminosite1	Variation	Catégorie de Variati
Chaouia - Ouardigha	lundi 1 janvier 2024	0,2495122047061 52	35,6165347750878		2024	1	Hiver			-0,0367131603794412	Baisse

Figure 2: Cleaned and enriched dataset in Power BI after column transformations.

IV. Data Visualisation

1. General NDVI Visualisation

Figure 3: Power BI dashboard – General NDVI Visualisation page.

a. General Overview

The national average NDVI over 2010–2024 is **0.26**, indicating overall **sparse vegetation** across Morocco. The range between the maximum (0.66) and minimum (0.08) reflects **significant heterogeneity** in vegetation cover.

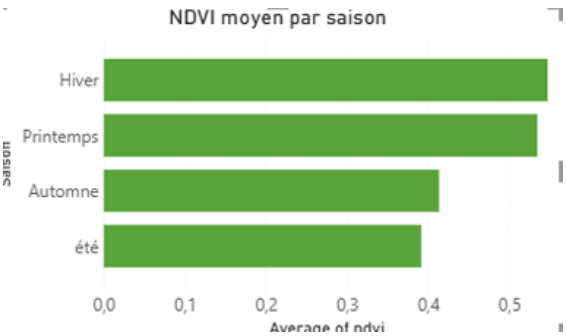
b. Regional Analysis

The *"Average NDVI by Region"* map shows a clear spatial gradient: NDVI decreases moving southward and eastward. Vegetation is most abundant in the **north and west**, and sparse in the **south and east**. The top/bottom-5 region charts confirm this:

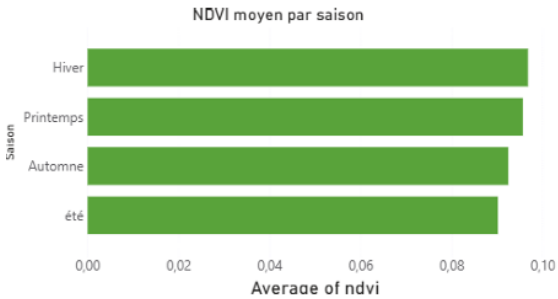
- **Highest NDVI variation (north/west):** Tanger-Tétouan, Gharb-Chrarda, Rabat-Salé-Zemmour, Grand Casablanca, Doukkala-Abda.
- **Lowest/negative variation (south/east):** Laâyoune-Boujdour, Guelmim-Es-Semara, Souss-Massa-Draâ, Oriental, Meknès-Tafilalet.

c. Seasonality Analysis

Northern and western regions display strong seasonality: NDVI peaks in **December–March** and reaches its minimum in **July**. Southern and eastern regions show almost no seasonal variation, with inter-season differences of 0.01–0.02 at most.



(a) Tanger-Tétouan (north): winter–summer difference: **0.11**.



(b) Laâyoune (south): winter–summer difference: **0.01**.

Figure 4: Average NDVI by season – northern vs. southern region.

The *"Average NDVI per Month"* chart confirms a national peak in **February–March** and a trough in **July–September**.

d. Annual NDVI Trend

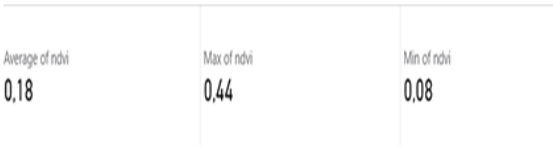
The *"Average NDVI per Year"* time series reveals a **downward trend**:

- **Maximum:** 0.28, in 2011 and 2018.
- **Minimum:** 0.23, in 2022 and 2024 – linked to Morocco's recent drought (verified in Section IV.2).

e. Conclusion

A strong geographic divide exists in Moroccan vegetation: average NDVI of **0.36** for north/west regions vs. **0.17** for south/east regions.

(a) North/West: Avg 0.36 · Max 0.66 · Min 0.17



(b) South/East: Avg 0.18 · Max 0.44 · Min 0.08

Figure 5: NDVI summary statistics by geographic zone (Power BI region slicers).

2. Correlations with Influencing Factors

a. Precipitation

The NDVI–precipitation correlation is $\mathbf{r = 0.90}$ — an exceptionally high coefficient in environmental analysis, reflecting a very strong positive linear relationship. As precipitation increases, NDVI increases directly. No outliers are present in the scatter plot.

Conclusion

NDVI is **strongly linked to precipitation**. Precipitation is the dominant driver of vegetation density and vigour in Morocco.

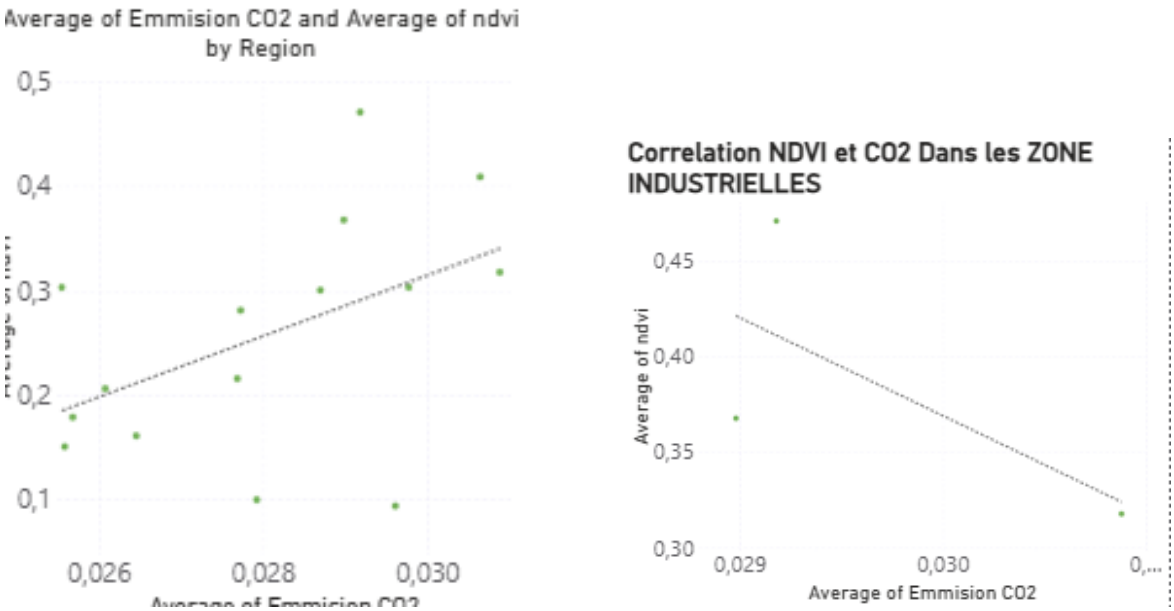
b. Temperature

The NDVI–temperature correlation is $\mathbf{r = -0.36}$, a moderate negative relationship. Higher temperatures intensify water stress in semi-arid regions, reducing vegetation vigour. Regions averaging 20–22°C exhibit lower NDVI; cooler regions (14–17°C) show higher values. Temperature influences vegetation, but is not the dominant factor.

c. Luminosity

The NDVI–luminosity correlation is $\mathbf{r = 0.86}$ — a very strong positive relationship. Regions receiving greater luminosity support denser, more active vegetation. The scatter plot shows a tight, well-aligned point cloud consistent across all regions.

d. CO_2 Emissions



(a) Overall correlation (r = 0.48) – confounded by industrial zones.

(b) Industrial zones only: negative correlation confirmed.

Figure 6: NDVI vs. CO_2 – full dataset vs. industrial zones only.

!! Misleading Statistic – Important Note

The raw correlation appears positive (r = 0.48). This is a **confounding effect**: Morocco's highest CO_2-emitting regions (Tanger, Casablanca, Rabat) also have the best climatic conditions for vegetation, acting as outliers that inflate the correlation. When restricted to industrial zones only, a **clearly negative relationship emerges**, confirming that CO_2 emissions negatively affect vegetation health.

e. Conclusion

Conclusion

Two factors **directly and positively** drive NDVI: **Precipitation** (r = 0.90) and **Luminosity** (r = 0.86). One factor exerts a **moderate negative** effect: **Temperature** (r = −0.36). CO_2 also negatively affects NDVI once geographic confounders are controlled for.
This explains the regional patterns in Section IV.1: northern and western regions achieve higher NDVI and stronger seasonality due to more precipitation, higher luminosity, and lower temperatures. Southern and eastern regions face persistently low precipitation and high temperatures, yielding low NDVI with negligible seasonal variation.

3. General Synthesis

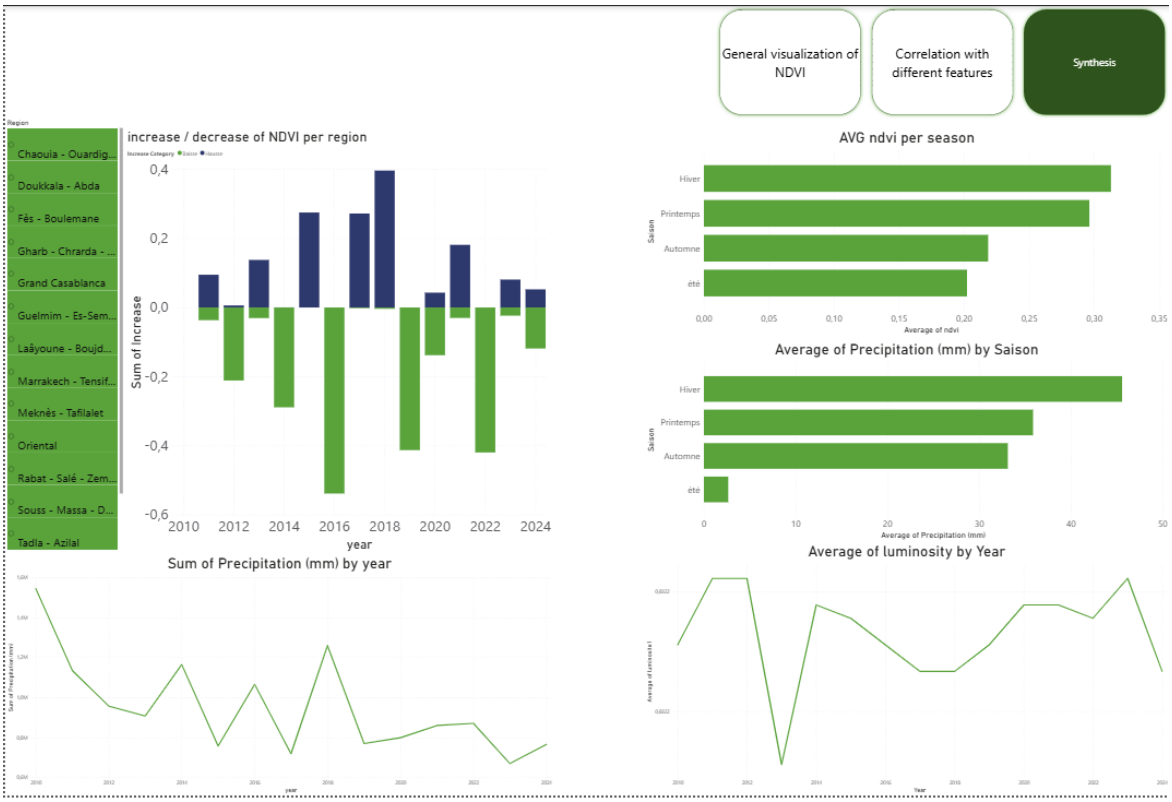


Figure 7: Power BI dashboard – Synthesis page.

- **Overall trend (2010–2024):** Morocco experienced more NDVI decreases than increases. Maximum decrease: $-\mathbf{0.6}$ in 2016; maximum increase: $+\mathbf{0.4}$ in 2018.
- **2016 drought confirmed:** In 2015, total national precipitation reached its minimum (757,769 mm), directly causing the record NDVI drop in 2016 – consistent with r = 0.90.
- **2022–2024 drought verified:** Precipitation data for those years confirms the NDVI decline hypothesis raised in Section IV.1.
- **Seasonal pattern:** Both *"Average NDVI by Season"* and *"Average Precipitation by Season"* follow the same ranking: Winter → Spring → Autumn → Summer.
- **Luminosity effect confirmed:** Years of peak luminosity consistently coincide with significant NDVI increases, validating r = 0.86.

V. Conclusion

This project provides a clear and structured picture of vegetation dynamics in Morocco from 2010 to 2024. By combining satellite-derived NDVI data with climatic variables — precipitation, temperature, luminosity, and CO_2 emissions — the analysis reveals both the spatial heterogeneity of vegetation cover and the key environmental drivers behind it.

The **north and west** of Morocco emerge as zones with the highest potential for sustained vegetation development, offering strategic opportunities for innovative agricultural projects, targeted investment, and the sustainable valorisation of natural resources. These findings provide a data-driven foundation for decision-making in environmental management and rural development policy.

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